MLA0205 - Fundamentals of Machine Learning.

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ASSIGNMENT - I

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Task1. Design of a Learning System.

System	Learning paradigm	Learning task	Experience (E)	Performance (P)
(a) loan default prediction	Supervised	Classification	Labelled repayment records	Accuracy, precision, recall.
(b) Customer segmentation	Unsupervised	Clustering	Unlabelled customer data	Silhouette score
(c) Credit-card spending	Supervised	Regression	Historical spending records	Mean Squared Error (MSE)
(d) ATM. cash-refill policy	Reinforcement	Control/ decision-making	Repeated trails and feedback	Reward, cash availability and transport cost.
(e) Fraud detection	Semi-supervised	Classification	Labelled and unlabelled transaction	Precision, recall and F1-score

Label noise analysis : System(e), fraud detection, can be particularly vulnerable because only 2% of transactions are labelled. Incorrect labels can mislead the model, especially when fraud cases are rare.

Task 2. Bayes' Theorem.

Given,

Prevalence = 2% = 0.02

Sensitivity = 95% = 0.95

Specificity = 90% = 0.90

False-positive rate = 1 − 0.90 = 0.10

Using Baye's Theorem,

$$P(D/+) = \frac{P(+/D)\,P(D)}{P(+/D)\,P(D) + P(+/\bar{D})\,P(\bar{D})}$$

$$= \frac{0.95\,(0.02)}{0.95(0.02) + 0.10(0.98)}$$

$$= \frac{0.019}{0.117} = 0.1624\ \%$$

Sensitivity measures the proportion of diseased people correctly identified. The probability of disease after a positive result also depends on prevalence and false positives.

Task 3. Joint Probability Distribution.

X	P(X, Y = 0)	P(X, Y = 1)
0	0.10	0.20
1	0.15	0.25
2	0.20	0.10

(a) Marginal probabilities,

For X, $P(X=0) = 0.10 + 0.20 = 0.30$

$P(X=1) = 0.15 + 0.25 = 0.40$

$P(X=2) = 0.20 + 0.10 = 0.30$

$$= \frac{0.10}{0.55} = 01818$$

(e) Independence,

For independent variables,

$$P(X=0, Y=0) = P(X=0)\,P(Y=0)$$

But,

$$0.10 \neq 0.30(0.45) = 0.135$$

$\therefore$ X and Y are not independent.

Task 4. Normal Distribution and Maximum Likelihood.

(a) Normal distribution probabilities,

Given,

$$X \sim N(50, 16), \quad \sigma = 4$$

$P(X>58)$: $\therefore z = \frac{58-50}{4} = 2$

$$P(X>58) = 1 - \phi(2)$$
$$= 1 - 0.9772 = 0.0228$$

For $P(46<X<54)$: $z_1 = \frac{46-50}{4} = -1$, $z_2 = \frac{54-50}{4} = 1$

$$P(46<X<54) = \phi(1) - \phi(-1)$$
$$= 0.8413 - 0.1587 = 0.6826$$

(b) Maximum-likelihood estimates,

Readings : 12, 15, 11, 14, 13

Mean = 13

Sum of squared deviations = 10

For Y : $P(Y=0) = 0.10 + 0.15 + 0.20 = 0.45$

$P(Y=1) = 0.20 + 0.25 + 0.10 = 0.55$

(b) Calculate $E[x]$, $Var(x)$ and $E[Y]$,

$$E[x] = \sum x\, P(X=x)$$

$$= 0(0.30) + 1(0.40) + 2(0.30) = 1$$

$$E[X^2] = 0^2(0.30) + 1^2(0.40) + 2^2(0.30) = 1.60$$

$$Var(X) = E[X^2] - (E[X])^2$$

$$= 1.60 - 1^2 = \underline{0.60}$$

$$E[Y] = 0(0.45) + 1(0.55) = \underline{0.55}$$

(d) Convariance and correlation coefficient,

$$E[XY] = 1(1)(0.25) + 2(1)(0.10) = 0.45$$

$$Cov(X,Y) = E[XY] - E[X]E[Y]$$

$$= 0.45 - (1)(0.55) = \underline{-0.10}$$

Since Y is binary,

$$Var(Y) = 0.55(0.45) = 0.2475$$

$$\rho = \frac{Cov(X,Y)}{\sqrt{Var(X) \cdot Var(Y)}}$$

$$= \frac{-0.10}{\sqrt{0.60(0.2475)}} = -0.2595$$

(c) Conditional probability,

$$P(X=2 \mid Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)}$$

ML variance : $\sigma^2_{ML} = \frac{10}{5} = 2$

Unbiased sample variance : $S^2 = \frac{10}{5-1} = 2.5$

Task 5. Entropy and Coding

Given, $P = (1/2, 1/4, 1/8, 1/8)$

Entropy, $H(X) = -\sum p_i \log_2 p_i$

$= -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + 2\left[\frac{1}{8}\log_2\frac{1}{8}\right]\right]$

$= 1.75$ bits

Uniform source entropy, $H_u = \log_2 4 = 2$ bits

Redundancy, $R = 2 - 1.75 = 0.25$ bits

Relative redundancy = 12.5 %.

For code $\{0, 10, 110, 111\}$,

$L = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$

$L = 1.75$ bits / symbol

The code is prefix-free and optimal for these probabilities.

Task 6. KL Divergence

Given, $P = (0.50, 0.25)$ $Q = (1/3, 1/3, 1/3)$

$D_{KL}(P||Q) = \sum P_i \log_2 \frac{P_i}{Q_i}$

$D_{KL}(P||Q) = 0.08496$ bits

Similarly, $D_{KL}(Q||P) = 0.08170$ bits

KL divergence is asymmetric, so reversing the distributions changes.

Task 7. Entropy and Mutual Information.

Given,

The joint probabilities 0.4, 0.1, 0.1, 0.4

$$H(A) = H(B) = 1 \text{ bit}$$

$$H(A,B) = -2(0.4 \log_2 0.4) - 2(0.1 \log_2 0.1)$$

$$H(A,B) = 1.7219 \text{ bits}$$

$$H(A|B) = 1.7219 - 1 = 0.7219 \text{ bits}$$

$$I(A,B) = 1 + 1 - 1.7219 = 0.2781 \text{ bits}$$

∴ B provides some useful information for predicting A, but it doesnot determine A completely.

Task 8. Bayes Decision Theory.

Given mean 2 and 6, common variance 4, and prior 0.6 and 0.4.

Threshold satisfies, $0.6p(x|\omega_1) = 0.4p(x|\omega_2)$

$$x^* = 4 + \ln \frac{0.6}{0.4}$$

$$x^* = 4.4055$$

Since $4.5 > 4.4055$, classify $x = 4.5$ as class ω_2,

Overall error,

$$P_e = 0.6[1 - \phi(1.203)] + 0.4\phi(-0.797)$$

$$P_e \approx 15.38\%.$$

Increasing the prior of ω_1 moves the threshold to right, favouring ω_1.

Task 9. Minimum-Risk Decision

Given posterior probalities 0.3 and 0.7.

Risk of declaring disease,

$$R(\alpha_1|x) = 0(0.3) + 8(0.7) = 5.6.$$

Risk of declaring healthy,

$$R(\alpha_2|x) = 50(0.3) + 0(0.7) = 15.$$

Minimum-risk decision, Decleare disease (α_1).

MAP instead decleas healthy because its posterior probability,

$$8(1-p) < 50p$$

$$p > \frac{8}{58} \approx 13.79\%$$

Task 10. Bias-Variance Trade-off

Generalisation gap = Validation MSE - Training MSE

Degree	Training MSE	Validation MSE	Gap
1	9.0	9.5	0.5
2	4.1	4.6	0.5
3	2.2	2.9	0.7
5	0.9	3.6	2.7
9	0.1	11.8	11.7

Best model : Degree 3, because it has lowest validation MSE.

Degree 1 underfits, while degree 9 severely overfits.

Two improvements for degree 9.

1. Apply Ridge or Lasso regularisation.
2. Reduce the polynomial degree.

Task 11. ML, MAP and Posterior Mean.

(a) 7 heads in 10 tosses,

$$\hat{\theta}_{ML} = \frac{7}{10} = 0.70$$

Posterior : Beta (9,5)

$$\hat{\theta}_{MAP} = \frac{9-1}{9+5-2} = 0.6667$$

$$E[\theta|D] = \frac{9}{14} = 0.6429$$

(b) 70 heads in 100 tosses,

$$\hat{\theta}_{ML} = \frac{70}{100} = 0.70$$

Posterior : Beta (72, 32)

$$\hat{\theta}_{MAP} = \frac{71}{102} = 0.6961$$

$$E[\theta|D] = \frac{72}{104} = 0.6923$$

With more observations, the influence of the prior decreases; and MAP and posterior mean approach the ML estimate.

Task 12. Hospital Sepsis Early-Warning System

(a) Design of learning system.

Task : Predict sepsis within six hours.

Performance : Sensitivity, specificity and predictive values.

Experience : Historical patient records and outcomes.

Target functions : Predict the probability of sepsis.

Inputs : Vital signs and laboratory results.

(b) Confusion matrix,

Total patients = 1000

Sepsis = 50

No sepsis = 950

TP = 50(0.90) = 45, FN = 5

TN = 950(0.80) = 760, FP = 190

	Alert	No alert
Sepsis		FN = 5
No Sepsis	FP = 45 FP = 190	TN = 760

(c) Predictive values,

$$PPV = \frac{45}{45+190} \times 100 = 19.15\%$$

$$NPV = \frac{760}{760+5} \times 100 = 99.35\%$$

(d) Total cost,

Cost of missed case = ₹100

false alarm = ₹5

Using the model,

5(100) + 190(5) = ₹1,450

Alerting everyone,

950(5) = ₹4,750

Never alerting,

50(100) = ₹5,000

The model has the lowest total cost.

(e) Deployment policy,

Use the model as a clinical decision - support tool with professional review.

Alert when, $5(1-p) < 100p$

$$\boxed{p > 4.76\%}$$

The threshold should be validated for patient safety and clinical requirements.